Lab Problem 8

Objective:

To implement a Naive Bayes classifier on the SMS Spam Collection dataset, apply Laplace smoothing to handle the zero-frequency problem, and evaluate its impact on classification performance.

Dataset : SMS Spam Collection Dataset (UCI Repository)

- The **SMS Spam Collection Dataset** contains SMS messages labeled as Spam or Ham. The data is converted into numerical features using the Bag-of-Words model.

Task 1: Data Preparation

- Download the dataset, preprocess the text (tokenization and stop word removal), convert it into Bag-of-Words features, and split into training and testing sets.

Task 2: Naive Bayes without Smoothing

- Compute prior and conditional probabilities, train the Naive Bayes classifier without smoothing, evaluate using accuracy, precision, recall, and F1-score, and identify zero-probability cases.

Task 3: Laplace Smoothing

- Apply Laplace smoothing to the conditional probabilities and recompute them.

Task 4: Naive Bayes with Smoothing

- Train the classifier using smoothed probabilities and evaluate using accuracy, precision, recall, and F1-score.

Task 5: Comparison and Analysis

- Compare both results, explain the zero-frequency problem, describe how smoothing resolves it, and conclude which method performs better.

Implementation Task

- Write a Python program to implement Naive Bayes classification, apply Laplace smoothing, evaluate performance using accuracy, precision, recall, and F1-score, and plot a bar graph comparing the performance before and after smoothing.